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Master's thesis

Rethinking Multilevel K-Way Graph Partitioning using an Objective Function that Considers the Distribution of Communication volume

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Abstract

Here come 3–6 sentences describing your thesis.

Preface

Here comes your preface, including acknowledgments and thanks.

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Acronyms

GP	Graph Partitioning
GPP	Graph Partitioning Problem
BFS	Breadth-First Search
SGP	Streaming Graph Partitioning
GGP	Graph Growing Partitioning
KL	Kernighan-Lin
GR	Greedy Refinement
MPI	Message Passing Interface

Chapter 1

Introduction

The demand for computational resources in modern scientific and engineering applications keeps increasing, making parallel computing an essential area of development. While processor performance continues to improve, the growth in single-core performance has slowed considerably since the end of Dennard scaling [21], making it impractical to tackle the largest problems on a single processor alone. Instead, it has become necessary to distribute these workloads across the many processors of a parallel computing system.

In order to compute large communicating tasks in parallel computing, many models have been proposed that represent them as graphs or hypergraphs [5, 6, 8, 18, 44]. The vertices correspond to units of computation and the edges encode data dependencies between them [18]. Efficiently distributing the computation then becomes a question of how to partition this graph among processors: balancing the workload evenly while minimizing the communication between them. This is also known as the *graph partitioning problem*. Although graph partitioning is known to be NP-complete, a variety of algorithms have been proposed that produce good quality solutions in practice by using heuristics [19, 24, 26, 27]. Beyond parallel computing, graph partitioning plays a critical role in numerous other application areas, including task scheduling, neural network simulations, operations research and more.

More than 20 years ago the scientific community already argued that the future of software development lies in concurrency, e.g [42]. As computational demands continue to grow, our programs and requirements consistently outpace hardware improvements. Sutter highlights the need for better parallel programming tools, given the inherent difficulty of developing parallel solutions. A decade later, on July 29, 2015, President Obama issued Executive Order 13702, establishing the National Strategic Computing Initiative (NSCI) to ensure continued United States leadership in high-performance computing [12], underscoring the importance of this field and the need for sustained research. Graph partitioning algorithms are central to making this possible.

Most modern graph partitioning algorithms minimize for *edge cut* (see Section 2.1.1), seeking to minimize the number of inter-partition edges. The appeal of this metric lies in its simplicity: it is easy to compute, and for the structured meshes that dominate scientific computing, it correlates reasonably well with actual communication costs [18]. However, edge cut has several well known shortcomings as a representation for communication [17, 18]. First, it over counts *communication volume* (see Section 2.1.1): a vertex with multiple edges crossing into the same partition needs to be communicated only once, yet each edge is counted separately when using edge cut. Second, the metric ignores data communication latency, the startup cost when sharing data between

partitions, which can dominate communication cost when the size of the communicated data is small. Third, and most relevant to this work, minimizing the *total* edge cut does not account for how communication is distributed across partitions. In a parallel application, performance is generally limited by the slowest processor [18], meaning that even if balanced globally, some processors might bear a greater load than others, creating bottlenecks.

Karypis and Moulitsas highlighted the need for partitioning algorithms that account for heterogeneous architectures and proposed using *communication volume* as an optimization objective instead of edge cut [33]. This communication volume objective was subsequently added into the METIS software package. However, like the edge cut objective, METIS focuses on the total communication volume without considering how volume is distributed among partitions. Regardless of the architecture, focusing only on the total volume of a partitioned graph may lead to bottlenecks in some partitions.

Efficiently distributing computation across processors is central to high-performance computing, and graph partitioning is the standard technique for achieving this. In *Graph Partitioning and Parallel Solvers: Has the Emperor No Clothes?* [17], Hendrickson concludes that one of the key needs is to develop a more accurate metric for communication cost. Researchers have proposed alternative minimization objectives that target per-processor communication metrics rather than global totals. An example is UMPa [8], which minimizes the maximum data sent by any single processor in a distributed memory setting.

1.1 Thesis Goal

As Hendrickson noted, minimization objectives of this kind are harder to optimize than the standard edge cut, which partly explains their limited adoption. Despite these difficulties, in this thesis, we present a new objective called *NVOL*. NVOL minimizes the *per-partition* maximum volume, aiming to reduce uneven communication loads. Using the METIS software as our baseline multilevel k-way partitioner, we implement this new partition-aware objective and address two questions: Can a partition-aware objective produce better balanced communication loads than a global one? Can the performance of a parallel application be improved in practice by using a partition-aware objective?

Our new objective, NVOL, will gather information about how every partition's communication volume is affected by a vertex changing partition. NVOL is able to correctly evaluate how much partition a will change if a vertex v is moved to another partition b. Using this information, NVOL prioritizes moves that improve the per-partition maximum. No move that increases the maximum volume held by a single partition is accepted. This means that in some cases NVOL might even accept moves that increase the total communication volume, if it decreases the per-partition maximum. The goal is to provide more balanced partitions, avoiding to overload some for the sake of a better total communication volume.

1.2 Thesis Structure

Chapter 2: *Background*

> Chapter 2 covers some of the fundamental topics in graph partitioning. It also shows some of the algorithms used today and some applications of graph partitioning algorithms.

Chapter 3: *Multilevel K-Way Graph Partitioning*

Chapter 3 discusses the Multilevel K-way graph partitioning algorithm used to develop NVOL. This approach achieves a good balance between quality and run time, making it a popular option for large graphs.

Chapter 4:

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Chapter 5:

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Chapter 6:

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Chapter 7:

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Chapter 2

Background

This chapter gives a description of what the *graph partitioning problem* (GPP) is, as well as what objective functions are used to evaluate partitions. It also discusses some of the common practices in the field with some examples of concrete application areas for *graph partitioning* (GP).

2.1 Graph Partitioning Problem

The GPP refers to a group of problems that deal with distributing the vertices of a graph into groups, clusters or communities. The goal is to group together vertices that are similar or that share many edges between them to minimize the number of edges between groups. Graphs may also be directed; however, directed graphs are outside the scope of this thesis, and we consider only the undirected case.

Consider an undirected graph $G = (V, E)$ with $|V| = n$ vertices and $|E| = m$ edges. The goal is to partition V into k disjoint subsets $V_1, V_2, \ldots, V_k$ such that $V_i \cap V_j = \emptyset$ for $i \neq j$ and $\bigcup_i V_i = V$ [26]. The weight of the graph is $W = \sum_{v \in V} w(v)$, where $w(v)$ is the weight of vertex v (vertices can also have multiple weights, but in this thesis we only consider graphs with a single vertex weight). Note that a graph with unit weights will have a weight equal to the number of vertices $W = n$. A balance constraint requires that all subsets have approximately equal weight. The partition weight equals the weight of each vertex it contains, $w(V_i) = \sum_{v \in V_i} w(v)$. Using an imbalance parameter ϵ, this constraint requires that each subset satisfies

$$w(V_i) \leq \left\lceil \frac{W}{k} \right\rceil \cdot (1 + \epsilon), \quad i = 1, 2, \ldots, k.$$

The parameter ϵ represents the fractional excess weight a subset may have beyond the perfectly balanced weight $\lceil W/k \rceil$. When $\epsilon = 0$, the partition must be perfectly balanced [4]. The goal of GPP is to minimize a given objective function (see Section 2.1.1) while satisfying the balance constraint. GPP is NP-complete, meaning that for large graphs a solution can only be achieved through heuristics or approximation algorithms [16]. In GP, the most common practice is to use heuristics. Approximation algorithms, regardless of their high theoretical importance, either underperform compared to state-of-the-art GP solvers, or are too slow for large graphs [4].

Another common representation of computational tasks in GP is *hypergraphs*. A hypergraph is a generalization of a graph where an edge, also called a hyperedge, can connect any number of vertices [3]. Similarly, GP divides the vertices into partitions while trying to maintain roughly equal size for each. However, given this more complex

structure, utilizing hypergraphs requires more complex algorithms both in terms of implementation and running time. For the scope of this thesis, we will focus on standard GP; however, most of the techniques presented in this section (see Section 2.2) have been or can be applied to a solution based on hypergraphs as well.

2.1.1 Objective Functions

The most common objective in GPP is to minimize the total edge cut [4]. An edge $\{u, v\} \in E$ is said to be *cut* if its endpoints belong to different partitions, i.e., $u \in V_i$ and $v \in V_j$ with $i \neq j$. The total edge cut is then the sum of all such cut edges across partitions. This also applies for graphs with edge weights, where the total edge cut is then the sum of the edge weights of cut edges.

Another common objective is communication volume, which provides a more accurate measure of the data exchange between partitions of a communicating task. For a partition V_i, the outgoing communication volume is defined as $\text{comm}(V_i) = \sum_{v \in V_i} D(v)w(v)$, where $D(v)$ denotes the number of distinct partitions, other than the one containing v, in which v has at least one edge to another vertex. The communication volume is then multiplied with the vertex weight, $w(v)$, as that represents the outgoing communication volume.

Despite being a less precise measure for actual communication cost, edge cut has been widely adopted as the standard objective. Buluç et al. [4] attribute this to two factors: cut optimization appears to be easier in practice, and for graphs with high structural locality, the cut tends to perform similarly to other objectives. Additionally, alternative objectives complicate the use of multilevel approaches making them harder to implement.

2.1.2 Shortcomings of the Objective Functions

Even though edge cut is accepted as the standard, it comes with several shortcomings. Most notably, it does not distinguish between multiple edges connecting to the same neighboring partition. Consider Figure 2.1, where each color represents a partition. Vertex A has two edges, both leading to the blue partition. Calculating the edge cut, A contributes 2 to the total edge cut. The communication volume, however, is only 1, since both edges connect to the same partition. In general, edge cut can overestimate the actual communication burden, particularly for vertices with many neighbors in a single remote partition.

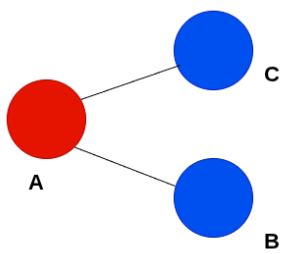

Figure 2.1: Vertex A has a communication volume of 1 but an edge cut of 2. Each color represents a partition.

The performance of a parallel application is generally limited by its slowest processor. This means that even if the computational work is balanced, the communication effort

may not be [4]. In such cases, it is more meaningful to minimize the maximum communication load incurred on a single processor rather than the total load across all processors. Neither the standard edge cut metric nor the total communication volume captures this type of objective.

Furthermore, on many architectures the time to send data, or messages, depends not on geographic distance but on the number of network switches the data must traverse. Although modern machines employ cut-through or wormhole routing to move individual messages quickly between distant processors, the communication network handles many messages simultaneously, and a message between distant processors occupies links that are unavailable to other traffic. Minimizing edge cut does not account for this locality of communication, and a partition that places frequently communicating processors far apart in the network topology can lead to significant message contention [4, 18].

Finally, the time to send a message on a parallel computer depends on both the latency (startup cost) and the message size. GP approaches typically aim to minimize the total communication volume but not the total number of messages sent. Depending on the machine architecture and problem size, message latencies can dominate message volume, making the number of distinct messages a relevant cost factor as well [4, 18]. A few big messages might be more efficient than many small ones.

Regardless of the objective function chosen, GP remains a fundamental tool across a wide range of domains. The following sections discuss some of the most used GP algorithms and some of their application areas.

2.2 Graph Partitioning Algorithms

In this section we will present some of the most popular GP algorithms used today. The goal is to show some of the differences in the approaches and to highlight their typical use cases. This will help the reader to understand why the multilevel partitioning approach is deemed the best option for large scale graphs. These are, however, only *some* of the existing GP algorithms, a more complete list can be found in [4, 7].

2.2.1 Recursive Bisection

A popular partitioning method is recursive bisection (RB). Instead of partitioning a graph directly into k subsets, RB first bisects the graph into two roughly equal parts using algorithms such as [1, 13, 29], and then recursively bisects each part until k partitions have been formed. When k is not a power of two, modified schemes are used [9, 19, 25, 39]. Although simple and widely used, RB has been shown to produce suboptimal k-way partitions [39], since each bisection is made independently without knowledge of the cuts that will follow at deeper levels of the recursion.

2.2.2 Spectral Bisection

Spectral Bisection was one of the first methods used to split a graph into two partitions and is still used today [4]. Its approach is to use the eigenvector, also known as the *Fiedler vector*, of the second smallest eigenvalue of the Laplacian matrix to deduce global information of the connectivity of the graph [14]. In order to form the two partitions, the median value $\overline{m}$ of the Fiedler vector is calculated, and every vertex with a value smaller or equal to $\overline{m}$ is assigned to one partition, and the rest to the other one.

The main computational cost of the spectral method lies in the calculation of the Fiedler vector. To optimize this, Barnard and Simon [1] have used a multilevel method

to produce an approximation of the Fiedler vector. The spectral method has also been extended to produce a partition for more than only two partitions and to support edge and vertex weights [19].

2.2.3 Graph Growing Partitioning

Another bisection approach is called graph growing partitioning (GGP) [25]. GGP is mainly based on breadth-first search (BFS). The core idea is to start from a random vertex and to use BFS to construct a partition. BFS is run until half the vertices are assigned to one partition, the remaining vertices are then assigned to the other one. The choice of starting vertex significantly affects the result, meaning that this algorithm tends to require multiple restarts to obtain a good solution. The GGP approach has also been extended to produce more than two partitions, called *Bubble framework* [11]. In this version k starting vertices are chosen, evenly distributed in the graph. Starting from these vertices, k BFS traversals are performed. Throughout, *Bubble* uses local search algorithms to balance the partitions in the graph. At the end of an iteration, a new starting vertex is chosen for all partitions. The new starting vertex is the vertex with the minimum total distance to all other vertices in the same partition. The algorithm is repeated until no new changes occur in the choice of starting vertices, or no improvement for the partitions is found. The major drawback of this approach is its high computational cost.

2.2.4 Streaming Graph Partitioning

Streaming data models, where input is processed on the fly rather than all at once, have gained popularity due to their small memory requirements. In the field of GP, the input graph is also passed as a stream, requiring much less space than the graph size [4]. Streaming graph partitioning (SGP) is also very fast, even faster than multilevel algorithms (see Section 2.2.5). However, these algorithms produce lower quality partitions compared to multilevel approaches. They are nonetheless useful in applications where speed is important, and an initial partition can be obtained from a stronger GP algorithm. For more details about SGP, we refer the reader to [36, 41, 43].

2.2.5 Multilevel Graph Partitioning

Multilevel GP is considered to be the most successful heuristic for partitioning large graphs [4, 25, 27]. Because of this, multilevel GP was chosen as the base on which to build our new objective function NVOL. In short, the idea is that a graph is coarsened to a smaller graph. Once the graph is at its smallest, an initial partition is performed, and then it is iteratively uncoarsened to the original size. During the uncoarsening, heuristics are applied to improve the partitions. The intuition behind multilevel GP is that it is easy to run intensive partitioning algorithms on the small graph without affecting the total runtime. Additionally, starting from an already good solution, the refinement stage is not costly to perform and can quickly improve the partitions. Chapter 3 will explain in detail the algorithm, providing the necessary background for understanding how NVOL is integrated into the framework.

2.3 Applications of Graph Partitioning

Graph partitioning arises in a broad variety of domains. This section highlights several of the most prominent application areas, though the list is by no means exhaustive.

2.3.1 Parallel Processing

Perhaps the most widespread application of GP is distributing computational work across the processors of a parallel machine [30]. GP is the standard approach in scientific computing applications such as sparse direct and iterative solvers. When the problem domain remains fixed throughout the computation, the partitioning can be performed once at the outset; this is known as static partitioning [4]. Other common applications in parallel processing include eigenvalue computations [2], breadth-first search [3], and algorithms for PageRank and connected components [38].

2.3.2 Complex Networks

GP is also relatively popular in the field of complex networks [4]. Complex networks, in this context, refer to weighted graphs with nontrivial structure arising from real world or modelling processes [35]. The objectives applied in this field are usually domain specific, making it hard to develop optimization algorithms.

Several application domains illustrate this breadth. In power grids, disturbances and cascading failures may lead to blackouts. GP can be used to isolate self-sufficient islands to prevent propagation of an outage [31]. Cut based objectives are sometimes extended to enhance the robustness of the system [32].

Geographically embedded networks have experienced a growth in requirements with the advances in location-aware devices (such as GPS). It is now vital that algorithms be able to analyze networks like these extremely fast. In these networks the vertices commonly represent entities in the real world tied to geographic places. The cut-based approach therefore needs to be reinforced with spatial contiguity constraints [4, 32].

Many biological systems can be modeled as graphs [4], protein-protein interaction or gene co-expression graphs, for example. Partitioning serves multiple purposes ranging from data reduction to the detection of biological processes. More about this field can be found in [22].

In social networks, community detection is one of the most popular topics. In contrast to traditional GPP, the number of subgroups or partitions is not known a priori [4]. Apart from this, GP has contributed greatly to the development of community detection algorithms [15, 34].

2.3.3 VLSI Physical Design

VLSI circuit design has historically been one of the primary application areas for graph partitioning [4]. The objective is to decompose a circuit into smaller subcircuits, ranging from gate arrays to complete integrated circuits, while minimizing the total interconnection cost, typically measured as the weight of connections crossing partition boundaries. In this formulation, nodes represent cells (i.e., basic logic or functional units such as gates) and edges represent wires. Scalability is essential, as circuits can contain millions of modules and partitioning constitutes a bottleneck in the overall design flow. Beyond the basic cut objective, practical VLSI partitioning must also respect domain specific constraints such as I/O placement, groups of cells that are required to remain in

the same block, and upper bounds on the cut size between block pairs. Further details can be found in [10, 23].

2.3.4 Neural Networks

Graph neural networks (GNNs) extend deep learning to graph structured data, enabling the processing of irregular domains that traditional neural network architectures cannot readily handle. However, as graph sizes grow, training GNNs becomes increasingly expensive due to rising communication overhead, particularly in distributed or multi-device settings. Graph partitioning offers a natural approach to this scalability challenge by distributing subgraphs across devices while minimizing inter-device communication.

A recent example is CaPGNN [40], a parallel full batch GNN training framework designed for multi-GPU servers. CaPGNN combines resource aware graph partitioning with an adaptive caching mechanism that exploits both CPU and GPU memory to reduce redundant data transfers. By taking into account the heterogeneous computational and communication capacities of individual GPUs, the framework achieves substantial reductions in training time with minimal impact on model accuracy. While this is only one instance of a broader trend, it illustrates how classical graph partitioning techniques are finding renewed relevance in the context of modern machine learning workloads.

Another recent example is ParDNN [37], an automatic, generic, and non-intrusive partitioning strategy for Deep Neural Networks represented as graphs. Like CaPGNN, this strategy has the goal of minimizing the training time of the model without affecting accuracy. Through their experiments, Qararyah et al. show significant improvements compared to related work in efficiency, superlinear scaling of batch size and training throughput.

Chapter 3

Multilevel K-Way Graph Partitioning

This chapter presents the Multilevel K-way Graph Partitioning algorithm which is one of the most widespread GP algorithms. Each section will explain in detail the phases of the multilevel approach with special focus on the techniques used in the partitioner used in this thesis.

3.1 Preliminaries

The overall structure of a multilevel k-way graph partitioning algorithm is fairly straightforward. The process begins by coarsening the input graph until it is reduced to a size of only a few thousand vertices. At this coarsest level, an initial k-way partitioning is computed using a graph growing algorithm (see Section 2.2.3) [25]. The graph is then uncoarsened back to its original size in successive steps. At each level of uncoarsening, the partitioning is refined to improve quality by exploiting the increased degrees of freedom available at the finer level. Karypis and Kumar demonstrate in [26] that this multilevel approach yields high quality partitions, often comparable to, or better than, those produced by recursive bisection [25].

In Figure 3.1, this process is illustrated, showing how an initial graph G_0 is coarsened until it reaches G_4, which is smaller in size. At this level, an initial partition is performed on the coarsest graph. After that, the graph is step by step projected back to its original size, refining the partition at each level in order to minimize the chosen objective function [28].

The multilevel partitioning problem uses the same definitions as in Section 2.1. Additionally, we represent a k-way partition through a vector P. The vector has a length of n, where n is the number of vertices in the graph. Each element of P is an integer between 1 and k, indicating the partition the vertex belongs to. Specifically, a vertex $v \in V$ assigned to partition a will have $P[v] = a$. The next sections will now explain how each phase works and how the partitions are formed.

3.2 Coarsening Phase

During the coarsening phase, a sequence of increasingly smaller graphs is constructed, maintaining a similar structure to the original graph with fewer degrees of freedom. The most common approach is to aggregate groups of vertices into a single coarse vertex through matching [4, 25]. Formally, the idea is to contract pairs of matched vertices, where we define a matching as a set of edges no two of which share a vertex [4, 25]. Generally, a matching is considered good if it contains many edges. To achieve

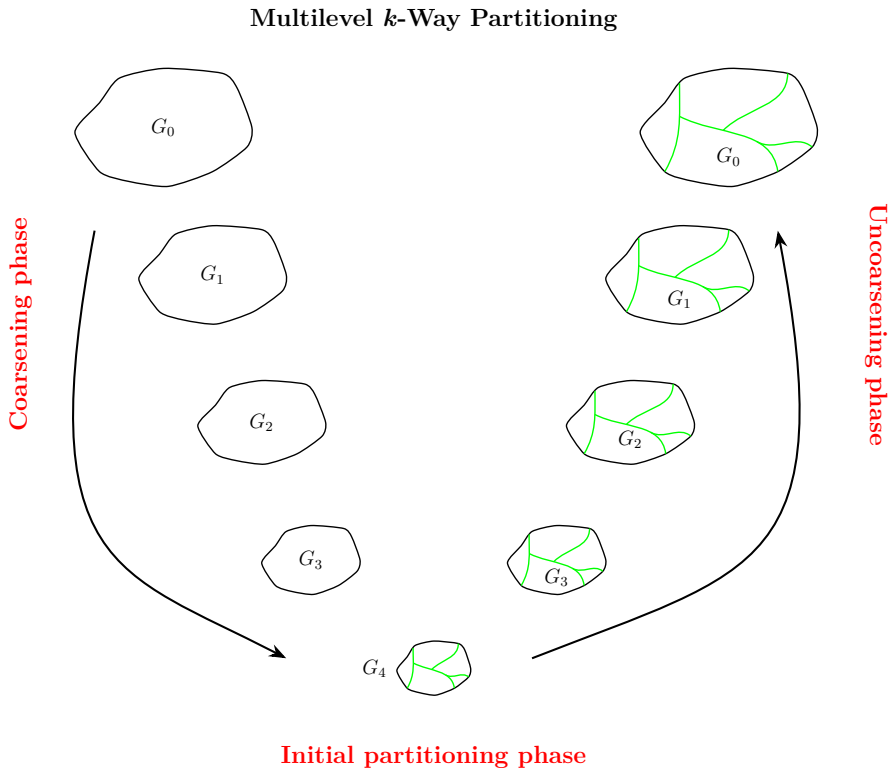

Figure 3.1: The phases of the multilevel k-way partitioning algorithm. During the coarsening phase, the vertices are contracted together and the graph size is successively decreased; during the initial partitioning phase, a k-way partition is performed on the smallest graph; and during the uncoarsening phase, the partitions are successively refined and projected back to a larger graph.

this, *maximal matchings* are used. A matching is maximal if any edge in the graph that is not contained in the matching has at least one of its endpoints matched [25]. The following subsection describes in detail the implementation of Sorted Heavy Edge Matching (SHEM), which is the matching algorithm employed by the partitioner used in this thesis. For further reading on other coarsening algorithms, we refer the reader to [4, 25].

Once a matching is performed the coarse vertex has a weight equal to the sum of the weights of all the vertices it contains. Additionally, the coarse vertex receives all the edges of the vertices it contains. Vertices that are not matched are simply copied over to the smaller graph. This process is repeated until we reach a graph G_m with a small enough number of vertices [24].

For a given graph $G_i = (V_i, E_i)$, a subset of vertices $V_i^v \subseteq V_i$ is merged into a coarse vertex $v \in V_{i+1}$ for the next coarser graph G_{i+1}. The coarse vertex v has weight $w(v) = \sum_{u \in V_i^v} w(u)$. This ensures that the total vertex weight is preserved across coarsening levels. To preserve connectivity, the new vertex v receives the union of the edges of the vertices in V_i^v. For any neighboring vertex u, the edge weight is $w(\{v, u\}) = \sum_{z \in V_i^v} w(\{z, u\})$. This aggregation strategy preserves structural and weight-related properties of the original graph, thereby enabling efficient and balanced processing at coarser levels [26].

3.2.1 Sorted Heavy Edge Matching

SHEM begins by generating a random permutation of the graph's vertices, which is then sorted based on vertex degrees. Each vertex gets assigned a bucket based on the square of its degree, sorted from lowest to highest. This ordering ensures that low-degree vertices, which have fewer opportunities for matching, are prioritized and thus more likely to be successfully matched. Delaying them would increase the chance of remaining unmatched. Once the vertices are sorted, they are traversed in order. For each unmatched vertex u, the algorithm selects the unmatched neighboring vertex v connected to u via the heaviest incident edge. The edge $\{u, v\}$ is then added to the maximal matching M_i, and both u and v are marked as matched. If no suitable neighbor exists, the vertex u remains unmatched. This process continues until all vertices have been processed.

The greedy nature of SHEM tends to preserve strong edge connections in the coarser graph, contributing to a significant reduction in total edge weight and, consequently, in the number of edge cuts [25, 27]. As shown in [24] by Karypis and Kumar, since the coarser graph has lower edge weight, it also has a lower edge cut.

3.3 Initial Partitioning Phase

Once the graph has been coarsened to the smallest level, the initial partitioning is performed. The goal of this stage is to produce k roughly equal partitions before further refinement. Several partitioning algorithms can be used here, including spectral partitioning [19], graph growing partitioning [25], and multilevel recursive bisection [39], among others. Since the graph is relatively small at this stage, the partitioning is fast. The partitioner we used, METIS, uses a multilevel recursive bisection approach. It is also worth mentioning that METIS uses different bisection algorithms based on the graph type. Using the definition in Section 2.1 we use GGP as the default bisection algorithm.

The a multilevel recursive bisection approach takes the coarse graph and bisects it using GGP. The resulting two subgraphs are then each bisected again. This process is repeated recursively until the desired number of partitions is reached. When the number of partitions is not a power of 2, the algorithm will keep bisecting until one graph has 3 partition remaining. At this level only one of the produced subgraphs will be further bisected. Consider Figure 3.2 where graph G is to be partitioned into 7 groups. At the first level, G_0 has 4 partitions remaining while G_1 has 3. Since G_1 has only 3 partitions remaining, only G_4 is bisected further. The left side (G_0), on the other hand, continues bisecting until all partitions have been produced.

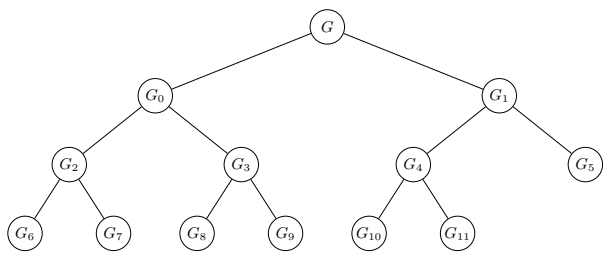

Figure 3.2: The process of bisecting a graph into 7 partitions.

During each bisecting step, GGP is used. As it was described in Section 2.2.3, this bisection approach bases itself on BFS. Starting from a random seed it tries to group together vertices connected to each other. At every step it checks that both partitions maintain a weight threshold. Once a partition is computed the graph is further refined

using the refinement algorithm proposed by Fiduccia and Mattheyses (FM) in [13]. The FM refinement algorithm goes through the boundary vertices of the graph, vertices that share at least one edge with a vertex in the other partition. For each step the vertex move that yields the greatest reduction in edge cut (see Section 2.1.1) is selected. In order to avoid a subpar solution, FM allows a limited number of moves that worsen the edge cut. This process is repeated over the boundary vertices several times, each time keeping the partition with the lowest edge cut.

It is worth noting that also for our volume based objective we have made use of this approach based on edge cut for the initial partition. There are a few reasons for this. Firstly, the initial partition is going to be heavily refined later during the uncoarsening phase, making the quality of the initial partition not as important. Additionally algorithms like FM are fast and work well with edge cut, optimizing for another objective would make this phase significantly slower. All of these reasons make it reasonable to use edge cut as an objective for this first partitioning.

3.4 Uncoarsening Phase

The final step is the uncoarsening and refinement phase, during which the graph is gradually projected back to its original size. The coarsest graph G_m is successively expanded through finer representations $G_{m-1}, G_{m-2}, \ldots, G_1$. At each projection from G_{i+1} to G_i, every vertex contained within a coarse vertex v inherits the partition assignment that v held in G_{i+1}. Once all coarse vertices at a given level have been expanded, the resulting partition is improved using local refinement heuristics.

Depending on the objective function chosen, the refinement heuristics will be different. In the rest of this section, we present the standard approach used by the partitioner we use, METIS, based on edge cut. The edge cut based refinement is simpler to understand, and will help the reader to get a better idea of how refinement works. In Chapter 4 and Chapter 5 we will discuss the communication volume objective and present our new and expanded objective NVOL.

METIS employs a variant of the Kernighan-Lin (KL) algorithm [29] to refine the graph. The KL algorithm iteratively swaps vertices among partitions of a bisection to minimize edge cut, continuing until no further improvement is possible and a local minimum is reached [26]. Refining a k-way partition is significantly more complex since each vertex has $k - 1$ possible moves as opposed to just one in a bisection[26]. This substantially enlarges the optimization space, forcing the algorithm to calculate the gain in each possible combination, thus increasing the computational complexity. The gain represents the change in total edge cut resulting from relocating a vertex to a new partition.

3.4.1 Projection and preparation before refinement

At every level of uncoarsening the graph is projected to a finer level. All the coarse vertices are split open, assigning each finer vertex to the same partition the coarse vertex had. Once the graph is projected up a level it undergoes a preparation phase before actual refinement. The new finer graph contains new vertices and edges, requiring a full recomputation of the underlying structures describing the graph. During this phase the whole graph is therefore traversed in order to get accurate information about the new connections in the graph. Additionally, the boundary set is recomputed, making sure to account for new boundary vertices.

The boundary set defines the list of vertices visited during refinement. So, to further increase efficiency METIS filters this set by more than wether a vertex is on the boundary or not. As the next section will show, a move is accepted only if it leads to positive or zero gain. Therefore, METIS chooses to ignore boundary vertices with negative gain.

In order to express gain for edge cut we need to introduce two more definitions. Any vertex has an internal and an external degree, which define the sum of the edge weights the vertex has toward different partitions. The internal degree is the sum of the weights of edges from a vertex v to other vertices in the same partition as v. We denote the internal degree of a vertex v with $ID[v]$. The external degree is the sum of the weights of edges from v to vertices in partitions other than its own. We denote the external degree of v as $ED[v]$. To simplify calculations during refinement we also specify the external degree towards a specific partition a as $ED_a[v]$.

A vertex v is then added to the boundary set if:

$$ED[v] - ID[v] \geq 0.$$

The specific external degree of a vertex will always be smaller or equal to the total external degree. Using Equation 3.1 we see that the gain will be positive only if the specific external degree is greater than the internal degree. Consequently, if the total external degree is already less than the internal degree no move will possibly lead to positive gain.

To optimize refinement even more, the boundary set is ordered by the potential gain a vertex has. Since the actual gain needs to be calculated for every possible move a vertex has, METIS makes an estimate for this potential gain. For a vertex v this potential gain is expressed as:

$$\frac{ED[v]}{\sqrt{|N_v|}} - ID[v]$$

where $|N_v|$ is the number of unique partitions v is connected to. This gives a normalized result of what the probable gain is. Using this ordering increases the chance of performing moves with high gain early.

3.4.2 Greedy Refinement

Karypis and Kumar propose an improved version of the k-way refinement KL algorithm in [26] referred to as Greedy Refinement (GR). This algorithm achieves high quality partitions while maintaining a computational complexity that is independent of the number of partitions k. Traditional KL-based refinement algorithms implement a lookahead strategy, which permits the movement of clusters of vertices across partition boundaries [26]. However, within the context of a multilevel partitioning framework, the benefit of this mechanism diminishes. Since each vertex in a coarser graph G_i represents a cluster of vertices from a finer graph G_{i-1}, moving a single vertex in G_i already corresponds to relocating a group of vertices in the original graph. Consequently, the lookahead mechanism becomes redundant. GR eliminates the lookahead step, thereby simplifying the refinement process and infrastructure presented in [20]. This simplification contributes to both the reduced computational complexity and the practical efficiency of the GR algorithm.

The GR algorithm proceeds through multiple iterations, where each iteration involves visiting all boundary vertices. For each vertex, the algorithm determines whether relocating it to a neighboring partition would improve the total edge cut. GR might in some cases also choose moves that do not change the edge cut, but that lead to an

improved balance in the graph. Any move must satisfy the balancing conditions [26]. The updated partitioning serves as the input for the next iteration.

For a graph $G_i = (V_i, E_i)$ and its partitioning vector P_i, each vertex $v \in V_i$ has a set of neighboring partitions N_v. That is, the union of the partitions that the vertices adjacent to v belong to. We denote the set of adjacent vertices to v as V_v^{adj}, we can now define: $N_v = \bigcup_{u \in V_v^{\text{adj}}} P_i[u]$ [26]. If all neighbors of v are in the same partition as v, then v is considered an *interior vertex*, and $N_v = \emptyset$. Conversely, if v has neighbors in different partitions, then $N_v \neq \emptyset$, and the cardinality of N_v can be as large as V_v^{adj} (in the case where each neighbor belongs to a distinct partition). During refinement, a vertex v can move to any of the partitions present in N_v.

For all boundary vertices we consider how a move to a neighbor partition (N_v) affects the total edge cut of the graph. For a boundary vertex v, moving to a neighbor partition $a \in N_v$ the gain is defined as:

$$\text{gain} = ED_a[v] - ID[v]. \tag{3.1}$$

GR will iterate over all partitions in N_v looking for any move that yields a positive gain. The move yielding the highest gain is then chosen as destination.

To maintain balance, each move needs to either follow the balance constraints of the graph or to improve the balance of the graph. The weight boundaries are defined as follows:

$$W^{min} = \frac{W}{k(1+\epsilon)}$$

$$W^{max} = \frac{W}{k} \cdot (1+\epsilon)$$

meaning that any partition V_i, must be $W^{min} < w(V_i) < W^{max}$ also after a move [26]. A small side note on the notation for W^{min}, which is different than the one found in [26], is that we based ours on the source code of the partitioner. Their only difference lies in the new expression using the balance constraint ϵ instead of a constant like in [26]. Additionally, GR accepts moves that improve the load balance in the graph. A move improves balance if the destination partition weighs less than the source after the move [26]. This ensures that the destination will not be overloaded by the move, or at least that it would be just as overloaded as the source. For a vertex v we express it as:

$$\frac{w(\text{destination})}{k} < \frac{w(\text{source}) - w(v)}{k}.$$

After each accepted move the boundary set, neighbor partition sets, and the degrees of the affected vertices are updated. GR will go through the boundary set until all vertices in it have been visited. Note that whenever a move is made the set is affected and vertices may get added or deleted from it. Additionally it's also worth mentioning that moved vertices are not allowed to be moved again during the same refinement stage [26]. Once a full pass is completed the boundary set is computed again and the process repeats itself. By default in our partitioner this process will repeat itself 10 times with the exception of an early termination if the pass led to no improvement in edge cut or no vertices were moved. Based on their empirical results, Karypis and Kumar observed that GR typically converges within four to eight iterations [26]. This relatively fast convergence contributes to the overall efficiency of the multilevel partitioning approach.

Chapter 4

Communication Volume Calculation

This chapter will show how the change in communication volume is calculated. The chapter will begin by explaining how the total communication volume objective works, followed by a definition of how our new partition-aware objective, NVOL, works. This chapter will only focus on how gain is evaluated and implemented in the multilevel framework for communication volume. The next chapter will show how a vertex move is chosen using the results from this calculation.

The total communication volume objective seeks to improve the total communication volume of a partitioned graph. Even though this metric is more representative of actual communication than edge cut, it does not necessarily produce a well balanced result. The net change in total communication volume after a potential move might hide bottlenecks and overloaded partitions. In contrast, NVOL delivers specific information about how the volume of each individual partition is affected by a move. Since NVOL was implemented on top of the METIS framework we will first spend some time formalizing how METIS and other partitioners use total communication volume as an objective in Section 4.2. In Section 4.3 we present the logic behind NVOL, and how we are able to evaluate how partitions are affected by a potential move.

Symbol	**Description**
G	The graph
V	Set of all the vertices in the graph
E	Set of all edges in the graph
V_b	Set of all boundary vertices in the graph
V_i^{adj}	Set of all vertices incident on vertex i
N_i	Set of neighboring partitions of i
N_{iv}	Set of partitions neighboring both i and v
N_{i-v}	Set of partitions only neighboring i
$w(i)$	Vertex weight of vertex i
$P[i]$	Partition of vertex i
$C_i[n]$	Number of edges a vertex i has to a partition n
G_i	Gain vector of a vertex i
SG_i	Gain table of a vertex i

Table 4.1: Notation used in Chapter 4.

4.1 Basic definitions

To facilitate understanding of the implementation (see Section 4.5), we begin by introducing a formal definition of communication volume. Let $G = (V, E)$ be a graph, and let P be the partition vector, where $P[v]$ is the partition a vertex v is assigned to. A vertex u is defined as a neighbor of v if there exists an edge $\{u, v\} \in E$. Any vertex adjacent to at least one vertex belonging to a different partition is said to lie on the boundary. We denote the set of all boundary vertices by $V_b \subseteq V$. Formally, a vertex $v \in V_b$ if there exists at least one neighbor $u \in V$ such that $\{v, u\} \in E$ and $P[v] \neq P[u]$.

For each boundary vertex $v \in V_b$, let N_v denote the set of distinct partitions (excluding $P[v]$) hosting its neighboring vertices. The number of partitions contained in N_v, defined as $|N_v|$, is the outgoing communication volume of vertex v. Note that the number of edges incident to v is not directly relevant for this measure, only the number of distinct neighboring partitions. In Figure 4.1, vertex A has a communication volume of 2, as it is connected to vertices in two distinct partitions. In contrast, Figure 4.2 illustrates a case where vertex A has two edges, both connecting to vertices within the same partition. Consequently, the communication volume of vertex A is 1, since only a single distinct external partition is involved.

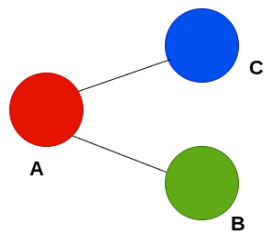

Figure 4.1: Vertex A has a communication volume of 2. Each color represents a partition.

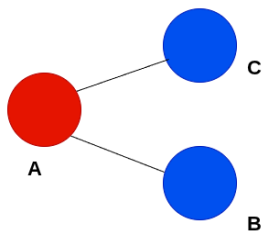

Figure 4.2: Vertex A has a communication volume of 1. Each color represents a partition.

The total communication volume of the partitioned graph is obtained by summing the outgoing volumes over all boundary vertices:

$$\text{total} \quad \text{volume} = \sum_{v \in V_b} |N_v| \, w(v) \tag{4.1}$$

Here, $w(v)$ denotes the weight associated with vertex v, which scales its contribution to the outgoing communication volume. Given the structure of the multilevel approach, regardless of a unit vertex weight graph, the vertices might have weights during uncoarsening (see Section 3.4). In Figure 4.3, all vertices are assumed to have a unit weight. The boundary set is $V_b = \{A, B, C, D, E\}$. Note that vertex F is not part of this set, as it has no edges connecting to vertices in other partitions.

Applying Equation 4.1, the total communication volume is computed by summing the contributions of all vertices in the boundary set. Vertices A, C, and D each contribute a volume of 2, while vertices B and E each contribute a volume of 1. This results in a total communication volume of 8.

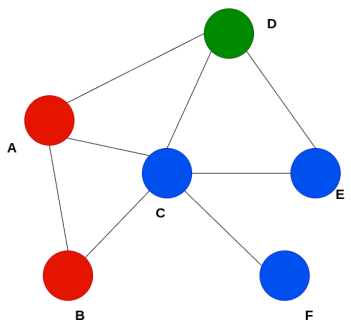

Figure 4.3: The total communication volume is 8.

4.2 Total gain calculation

To determine which vertices should be moved, and to which partitions, the gain need to be evaluated for any potential move. The gain represents the change in total communication volume resulting from relocating a vertex to a new partition. A positive gain indicates a reduction in total communication volume, whereas a negative gain corresponds to an increase.

Since communication only arises when an edge connects two vertices belonging to different partitions, a vertex move can only affect the communication volume of its neighboring partitions. For each boundary vertex $i \in V_b$, a partitioner needs to examine all its adjacent vertices $v \in V_i^{\text{adj}}$, where $V_i^{\text{adj}} \subseteq V$ denotes the set of vertices adjacent to i. When considering moving a boundary vertex i, the gain calculation depends on the relationship that i and each of its neighboring vertices have with other partitions. The effect on volume depends on which partitions they share. In particular, we define for every vertex i and every neighbor vertex v

$$N_{i-v} = N_i \setminus (N_v \cup P[v]) \quad \text{and} \quad N_{iv} = (N_i \cap N_v) \cup P[v] \tag{4.2}$$

representing partitions exclusive to i and partitions shared by i and v, respectively. In both cases, we add in $P[v]$ as this partition is shared even though it is not part of the N_v set. A vertex v has access to its own partition $P[v]$, meaning that $P[v]$ needs to be removed in N_{i-v} and added in N_{iv} to precisely determine shared partitions. Notice how both sets are subsets of $N_i = N_{i-v} \cup N_{iv}$.

As a concrete example, consider Figure 4.3 and suppose the candidate vertex is A, with neighbor B under consideration. Here, $N_A = \{\text{blue}, \text{green}\}$, while $N_B = \{\text{blue}\}$. Consequently, $N_{A-B} = \{\text{green}\}$, since green is the only partition connected to A but not to B, whereas $N_{AB} = \{\text{blue}\}$ as both vertices have at least one edge to the blue partition. Keeping A as the candidate vertex, consider now neighbor C with $N_C = \{\text{red}, \text{green}\}$. A and C have access to the same partitions, meaning that $N_{A-C} = \emptyset$ and $N_{AC} = \{\text{blue}, \text{green}\}$. This example highlights the need for adding $P[C]$ to the sets. The sets would have been $N_{A-C} = \{\text{blue}\}$, $N_{AC} = \{\text{green}\}$ if we had not accounted for $P[C]$. However, by simply viewing Figure 4.3 we can see that the blue partition is shared and not unique to A.

Let $C_i[n]$ denote the number of edges connecting i to partition n. In Figure 4.3, for example, $C_A[\text{blue}] = 1$, as A has only one edge connected to the blue partition, while $C_D[\text{blue}] = 2$, as D has two edges connected to the blue partition. Let G_i be a vector with a number of elements equal to the number of partitions, initialized at 0, where $G_i[n]$

represents the total gain obtained by moving vertex i to partition n. For a candidate vertex i being considered for a move, the gain contributions to all possible destination partitions are evaluated for each neighbor $v \in V_i^{\text{adj}}$ as follows:

Case 1. If $P[v] = P[i]$, then moving i to any partition in $k \in N_{i-v}$ creates a shared inter-partition connection for v, yielding

$$G_i[k] -= w(i) \tag{4.3}$$

Case 2. If $P[v] \neq P[i]$ and $C_v[P[i]] = 1$, then moving i to any partition $k \in N_{iv}$ eliminates an inter-partition connection for v, yielding

$$G_i[k] += w(i) \tag{4.4}$$

Case 3. If $P[v] \neq P[i]$ and $C_v[P[i]] > 1$, then moving i to any partition $k \in N_{i-v}$ introduces a new inter-partition connection for v, resulting in a gain contribution of

$$G_i[k] -= w(i) \tag{4.5}$$

Case 4. If $C_i[P[i]] = 0$, i.e., i has no internal degree, then moving i to any partition $k \in N_i$ yields an additional gain contribution of

$$G_i[k] += w(i) \tag{4.6}$$

For every adjacent boundary vertex v visited, we check each of the cases above for every partition in either N_{iv}, N_{i-v} or N_i depending on the case. These cases represent all possible changes in communication volume when moving a vertex. Algorithm 1 in Section 4.5 shows the flow of how the gain is calculated for a vertex i being considered moved.

4.3 Partition specific gain calculation

To analyze the communication volume at the partition level, we need to expand the strategy shown in Section 4.2. By only having the total gain for a move, information regarding how this gain affected each partition is missing.

In order to calculate the volume for a given partition n, consider a subset of boundary vertices where each vertex is part of the same partition. The communication volume associated with partition n is then given by

$$\text{partition} \quad \text{volume}[n] = \sum_{v \in V_b | P[v]=n} |N_v| \, w(v) \tag{4.7}$$

We introduce a table SG_i, where each row corresponds to a source partition and each column to a destination partition for vertex i. The entry $SG_i[n][k]$ represents the gain contribution to partition n when i is moved to partition k. In a graph with m partitions, a vertex i to partition n the total gain satisfies

$$\sum_{t=0}^{m-1} SG_i[t][n] = G_i[n] \tag{4.8}$$

i.e., the sum of each column in SG_i equals the corresponding entry in G_i. For a candidate vertex i being considered for a move, the partition-specific gain contributions to all possible destination partitions are evaluated for each neighbor $v \in V_i^{\text{adj}}$ as follows:

Case 1. If $P[v] = P[i]$, then moving i to any partition $k \in N_{i-v}$ removes a shared inter-partition connection. This affects the source partition:

$$SG_i[P[i]][k] -= w(i) \tag{4.9}$$

Case 2. If $P[v] \neq P[i]$ and $C_v[P[i]] = 1$, then moving i to any partition $k \in N_{iv}$ eliminates an inter-partition connection. This contributes to partition $P[v]$:

$$SG_i[P[v]][k] += w(i) \tag{4.10}$$

Case 3. If $P[v] \neq P[i]$ and $C_v[P[i]] > 1$, then moving i to any partition $k \in N_{i-v}$, introduces a new inter-partition connection. This contributes negatively to partition $P[v]$:

$$SG_i[P[v]][k] -= w(i) \tag{4.11}$$

Case 4. If $C_i[P[i]] = 0$, then moving i to any partition $k \in N_i$ yields an additional gain contribution for the destination partition:

$$SG_i[k][k] += w(i) \tag{4.12}$$

It is also necessary to account for how both the source and destination partitions are affected by a vertex move. Although such a move redistributes communication volume between partitions, it does not alter the total communication volume of the graph. As discussed previously, NVOL requires precise tracking of how each move impacts individual partitions. When vertex i is moved to partition k, this induces an additional gain contribution given by

$$SG_i[P[i]][k] += |N_i| \cdot w(i), \tag{4.13}$$
$$SG_i[k][k] -= |N_i| \cdot w(i). \tag{4.14}$$

This is referred to as a global term operation, as it reflects the overall redistribution of communication volume between the source and destination partitions of vertex i, rather than interactions with individual neighbor vertices. Equation 4.13 represents the volume leaving the source partition, while Equation 4.14 the volume the destination receives. Consequently, it is computed once per vertex move, in contrast to the neighbor specific gain contributions described earlier, which are evaluated separately for each adjacent vertex.

Apart from this distinction, both strategies follow the same case structure; the principal difference lies in how they store the gain information. Rather than recording only the magnitude of the change, NVOL also tracks which partition is affected.

4.4 Examples

To better understand the definitions of communication volume and gain, we present a sequence of illustrative examples. Each example highlights a particular configuration and demonstrates how gain contributions arise under different vertex moves.

4.4.1 Two Partition example

Consider the graph in Figure 4.4, consisting of five vertices with unit weights. The graph is partitioned into two subsets: the red partition containing vertices $\{A, B, C\}$, and the blue partition containing vertices $\{D, E\}$. Using Equation 4.1, the total communication volume is 4.

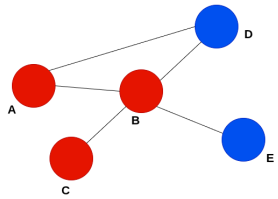

Figure 4.4: Graph with a volume of 4, each partition has a volume of 2.

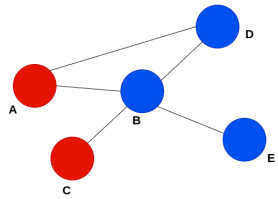

Figure 4.5: Graph with a volume of 4, each partition has a volume of 2 after the move.

Vertex A has one edge to the blue partition and contributes 1 to the communication volume. Vertex B has two edges incident to the blue partition, but contributes only 1 under the given definition. Similarly, vertices D and E each contribute 1. In contrast, vertex C contributes nothing, as all its incident edges are internal to its partition.

Now consider moving vertex B from the red partition to the blue partition, as shown in Figure 4.5. We calculate the gain using the global strategy (Section 4.2). Let B be the candidate vertex with neighbors $\{A, C, D, E\}$. Since there are two partitions, the gain vector G_B has size two.

Vertex *A*. Vertices A and B are in the same partition and both already have connections to the blue partition. Hence, the set N_{B-A} is empty and no gain contribution arises.

Vertex *C*. Vertex C lies in the red partition and has no edges to the blue partition. After moving B, C gains an inter-partition edge, increasing the communication volume. As shown in Case 1, this yields a negative gain:

$$G_B[\text{blue}] -= 1.$$

Vertex *D*. Vertex D is in the blue partition meaning that a move to the blue partition would decrease its volume only if one edge were incident on the red partition. Since D has two, Case 3 does not apply, moving B does not change its contribution, hence no case applies.

Vertex *E*. Vertex E satisfies Case 2: it has exactly one connection to the red partition. After the move, this connection disappears, reducing the communication volume and yielding a positive gain:

$$G_B[\text{blue}] += 1.$$

Case 4 does not apply, as B has multiple connections within its original partition. Summing all contributions yields total gain 0, so the communication volume remains unchanged.

Partition-specific formulation. Using the formulation from Section 4.3, we obtain the same result. We use the table SG_B with size 2×1 (the source column will always be zeroed out), to store each volume change. As shown above only vertices C and E contribute:

$$SG_B[\text{red}][\text{blue}] -= 1, \quad SG_B[\text{blue}][\text{blue}] += 1.$$

Additionally, accounting for the direct effect of moving B via Equations 4.13 and 4.14:

$$SG_B[\text{red}][\text{blue}] += 1, \quad SG_B[\text{blue}][\text{blue}] -= 1.$$

The net effect is zero, confirming that neither total nor per-partition volume changes.

4.4.2 Three partition example

We now extend the example to three partitions. Consider the graph in Figure 4.6, where all vertices have unit weight. The partitions are: red $\{A, B\}$, blue $\{C, D\}$, and green $\{E\}$. The total communication volume is 8, with partition volumes 3, 3, and 2, respectively.

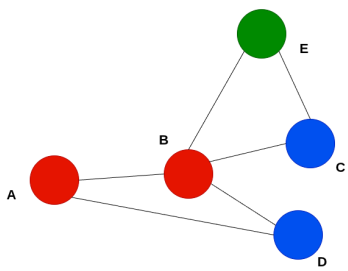

Figure 4.6: Graph with a total communication volume of 8.

Let B be the candidate vertex with neighbors $\{A, C, D, E\}$. Unlike the previous example, B may move to either the blue or green partition. Thus, G_B has size 3 and SG_B 3×2.

Vertex *A*. Vertex A is in the same partition as B. Since A is connected to the blue partition but not the green partition, Case 1 applies only for a move to green, introducing a new inter-partition edge:

$$G_B[\text{green}] -= 1.$$

Vertex *C*. Vertex C is outside the red partition and has exactly one edge to it. By Case 2, removing this edge reduces the communication volume. Since C is connected to both the blue and green partitions:

$$G_B[\text{blue}] += 1, \quad G_B[\text{green}] += 1.$$

Vertex *D*. Vertex D has two edges to the red partition. By Case 3, a gain change occurs only when the destination partition is not already adjacent to D. This holds for the green partition:

$$G_B[\text{green}] -= 1.$$

Vertex *E*. Vertex E is symmetrical to C, yielding:

$$G_B[\text{blue}] \mathrel{+}= 1, \quad G_B[\text{green}] \mathrel{+}= 1.$$

Summing contributions gives:

$$G_B = [0, 2, 0],$$

indicating that moving B to the blue partition reduces the total communication volume by 2, while moving it to the green partition leaves it unchanged.

Partition-specific formulation. The vector G_B alone does not capture how the volume is redistributed among individual partitions. Using Section 4.3, we compute SG_B as follows.

Vertex *A*. Vertex A follows Case 1for a move to the green partition, yielding:

$$SG_B[\text{red}][\text{green}] \mathrel{-}= 1$$

Vertex *C*. Vertex C follows Case 2 for both moves, yielding:

$$SG_B[\text{blue}][\text{blue}] \mathrel{+}= 1$$
$$SG_B[\text{blue}][\text{green}] \mathrel{+}= 1$$

Vertex *D*. Vertex D falls under Case 3 for the green partition, yielding:

$$SG_B[\text{blue}][\text{green}] \mathrel{-}= 1,$$

Vertex *E*. Vertex E is symmetrical to C, yielding:

$$SG_B[\text{green}][\text{blue}] \mathrel{+}= 1$$
$$SG_B[\text{green}][\text{green}] \mathrel{+}= 1$$

Additionally, accounting for the direct effect of moving B via Equations 4.13 and 4.14:

$$SG_B[\text{red}][\text{blue}] \mathrel{+}= 2, \quad SG_B[\text{red}][\text{green}] \mathrel{+}= 2,$$

$$SG_B[\text{blue}][\text{blue}] \mathrel{-}= 2, \quad SG_B[\text{green}][\text{green}] \mathrel{-}= 2.$$

The resulting table SG_B is:

$$\begin{array}{c} \\ \text{red} \\ \text{blue} \\ \text{green} \end{array} \begin{array}{c} \begin{array}{cc} \text{blue} & \text{green} \end{array} \\ \begin{pmatrix} 2 & 1 \\ -1 & 0 \\ 1 & -1 \end{pmatrix} \end{array}$$

Each column corresponds to a destination partition, and each row describes how a partition's volume changes. One can verify that the column sums recover G_B: the sums are 0 (the red partition is implied as no move occurs if the vertex stays in the same partition it already part of), $2 + (-1) + 1 = 2$ and $1 + 0 + (-1) = 0$, consistent with $G_B = [0, 2, 0]$. This formulation provides finer insight: although moving B to the green partition leaves the total volume unchanged, it redistributes volume across partitions. In particular, the blue partition experiences an increase in volume at the partition level.

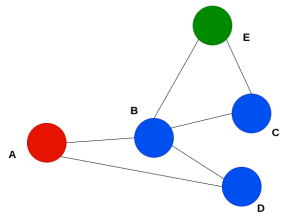

Figure 4.7: Graph has a volume of 6 after B is moved to the blue partition

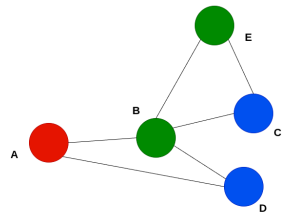

Figure 4.8: Graph has a volume of 8 after B is moved to the green partition

4.4.3 Case 4 example

None of the preceding examples illustrate a scenario in which Case 4 is relevant. To clarify how such a case may arise and how it affects the communication volume, consider Figure 4.9, where all vertices carry unit weight. The partition assignment is as follows: red $\{A, B, C, G\}$, blue $\{D, E\}$, and green $\{F\}$. The total communication volume is 8, with individual partition volumes of 3, 3, and 2, respectively. As in the previous example, let G_C have size 3 and SG_C size 3×3.

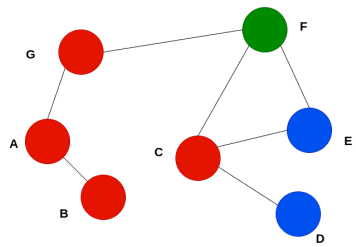

Figure 4.9: Graph with a total communication volume of 8.

The candidate vertex is C, which has no internal edges within the red partition. Despite the graph containing additional vertices, only the direct neighbors $\{D, E, F\}$ of C need to be visited in order to compute the gain. This illustrates how, even in large graphs, gain computation remains local to the immediate neighborhood of the moving vertex. For conciseness, both computation strategies are consolidated in this example, as they have been thoroughly demonstrated in earlier sections.

Vertex *D*. Vertex D satisfies Case 2 for a move to the blue partition, yielding:

$$\begin{aligned} G_C[\text{blue}] &+= 1 \\ SG_C[\text{blue}][\text{blue}] &+= 1 \end{aligned}$$

Vertex *E*. Vertex E satisfies Case 2 for all possible moves of C, yielding:

$$\begin{aligned} G_C[\text{blue}] &+= 1 \\ G_C[\text{green}] &+= 1 \\ SG_C[\text{blue}][\text{blue}] &+= 1 \\ SG_C[\text{blue}][\text{green}] &+= 1 \end{aligned}$$

Vertex *F*. Vertex F has two edges connecting it to the red partition; however, the set N_{C-F} is empty, meaning Case 3 does not apply and vertex F contributes no change to the gain

Case 4. Since vertex C has no internal edges, any move will reduce the volume of its source partition. Under normal circumstances, when the candidate vertex has at least one internal edge, moving it leaves the receiving partition's incoming volume unchanged through the candidate. In this case, however, it is necessary to increment the gain for every possible destination partition, yielding:

$$\begin{aligned} G_C[\text{blue}] &+= 1 \\ G_C[\text{green}] &+= 1 \\ SG_C[\text{blue}][\text{blue}] &+= 1 \\ SG_C[\text{green}][\text{green}] &+= 1 \end{aligned}$$

The resulting vector $G_C = [0\,3\,2]$. Accounting for the direct effect of relocating C via Equations **??** and **??**:

$$SG_C[\text{red}][\text{blue}] += 2, \quad SG_C[\text{red}][\text{green}] += 2,$$

$$SG_C[\text{blue}][\text{blue}] -= 2, \quad SG_C[\text{green}][\text{green}] -= 2.$$

The resulting table SG_C is:

$$\begin{array}{c} \\ \text{red} \\ \text{blue} \\ \text{green} \end{array} \begin{array}{c} \begin{array}{cc} \text{blue} & \text{green} \end{array} \\ \begin{pmatrix} 2 & 2 \\ 1 & 1 \\ 0 & -1 \end{pmatrix} \end{array}$$

Figures 4.10 and 4.11 show the resulting graphs after moving C to the blue and green partitions respectively. The reader can verify that the partition volumes in each case are consistent with the gains predicted by SG_C and G_C. Without accounting for Case 4, the gain calculation would be incomplete, potentially leading to incorrect volume estimates when a vertex has no internal edges.

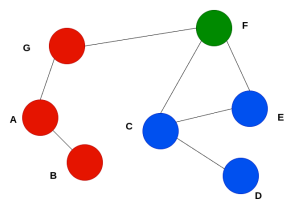

Figure 4.10: Graph has a volume of 5 after the move.

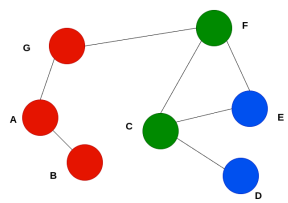

Figure 4.11: Graph has a volume of 6 after the move.

4.5 Implementation

We now describe how the calculation of communication volume change is integrated into the multilevel framework. As mentioned previously, we have used METIS as the base on which to develop NVOL. The following description bridges the formal definitions above and the implemented code.

4.5.1 Total gain implementation

In Section 3.4.1 the boundary set is computed using edge cut gain to filter and order it. When using total communication volume, we need to do the same thing using the calculations described in this chapter. Unlike the edge-cut metric, we cannot state for a given vertex whether moving it will lead to a positive gain or not. As stated in Section 3.4.1, we can deduce whether moving a vertex is guaranteed to give negative gain. This is not transferable to communication volume metrics.

With communication volume, it is necessary to visit all neighbors of a vertex to correctly evaluate the gains of possible moves. This makes the calculation difficult to integrate directly into the preparation phase before refinement. In METIS, this issue was resolved by adding the gain calculation as a final step after preparation. Once the graph has been projected and all underlying structures have been recomputed, a full scan of the graph is performed to evaluate the possible gains for every vertex.

Algorithm 1 Total gain calculation

```
nparts ← k                                          ▷ Graph with k partitions
for all i ∈ V do
    if |N_i| > 0 then                               ▷ Only visit vertices with neighboring partitions
        for all v ∈ V_i^adj do                      ▷ Visit all adjacent vertices of i
            for n = 0; n < nparts do
                if n ∈ N_i AND n ∈ N_v then
                    N_iv.add(n)                     ▷ Shared partitions between i and v (N_iv)
                else if n ∈ N_i AND n ∉ N_v then
                    N_{i−v}.add(n)                  ▷ Partitions only i is connected to (N_{i−v})
                end if
            end for
            if P[i] = P[v] then
                for all n ∈ N_{i−v} do
                    G[n] −= w(i)                    ▷ Case 1
                end for
            else if P[i] != P[v] AND C_v[P[i]] = 1 then
                for all n ∈ N_iv do
                    G[n] += w(i)                    ▷ Case 2
                end for
            else if P[i] != P[v] AND C_v[P[i]] > 1 then
                for all n ∈ N_{i−v} do
                    G[n] −= w(i)                    ▷ Case 3
                end for
            end if
        end for
        if C_i[P[i]] = 0 then
            for all n ∈ N_i do
                G[n] += w(i)                        ▷ Case 4
            end for
        end if
        BoundarySet.add(MAX(G_v))                   ▷ Order by highest possible gain
    end if
end for
```

4.5.2 Partition specific gain implementation

The implementation of the partition-specific gain calculation required a different approach, even though the logic in the inner loop is very similar. Before starting refinement, METIS calculates and stores the gain of all boundary vertices. However, NVOL requires significantly more memory: for a gain vector of 10 elements, NVOL's gain table contains 90 entries. For this reason, we opted for an on-the-fly approach, computing the gain table for each vertex as it is visited during refinement. Beyond the memory overhead, a precomputed gain table would also require an update procedure to maintain consistency after each move. As this is not directly relevant to the research questions of this thesis, we leave this as future work, discussed further in Chapter 7.

NVOL is implemented directly in the refinement stage. During preparation, it uses the same structure as Section 4.5.1, ordering vertices based on their total gain. Once refinement begins, the algorithm visits every vertex in the boundary set. For each vertex i under consideration, we run the algorithm below:

Algorithm 2 Partition specific gain calculation

```
nparts ← k                                          ▷ Graph with k partitions
for all v ∈ V_i^adj do                              ▷ Visit all adjacent vertices of i
    for n = 0; n < nparts do
        if n ∈ N_i AND n ∈ N_v then
            N_iv.add(n)                             ▷ Shared partitions between i and v (N_iv)
        else if n ∈ N_i AND n ∉ N_v then
            N_i−v.add(n)                            ▷ Partitions only i is connected to (N_i−v)
        end if
    end for
    if P[i] = P[v] then
        for all n ∈ N_i−v do
            SG[P[v]][n] −= w(i)                     ▷ Case 1
        end for
    else if P[i] != P[v] AND C_v[P[i]] = 1 then
        for all n ∈ N_iv do
            SG[P[v]][n] += w(i)                     ▷ Case 2
        end for
    else if P[i] != P[v] AND C_v[P[i]] > 1 then
        for all n ∈ N_i−v do
            SG[P[v]][n] −= w(i)                     ▷ Case 3
        end for
    end if
end for
if C_i[P[i]] = 0 then
    for all n ∈ N_i do
        SG[n][n] += w(i)                            ▷ Case 4
    end for
end if
```

Chapter 5

Refinement using communication volume

This chapter will discuss how the actual refinement is performed. Using the gains calculated in the previous chapter, we will explain how a vertex move is decided. This chapter will also compare how the choice is made using a total communication volume objective function and the partition-aware communication volume used in NVOL.

5.1 Refinement using total communication volume

After the preparation phase is completed, refinement starts and the partitioner visits all the vertices in the boundary set. For each vertex i, all the neighboring partitions of i are visited, checking the values of the G_i vector. A vertex move is chosen if it improves the total communication volume while maintaining the weight constraints, just like for GR in Section 3.4.2. The refinement algorithm will look for the move with the highest value in G_i that is allowed by the balance constraints. To prioritize high gain moves, METIS relaxes the balance constraint in proportion to the gain. Using a constant, called the fudge factor f, a move to partition p is allowed to exceed the maximum partition weight by $f \cdot G_i[p]$. This allows moves that improve the communication volume significantly at the cost of a few extra vertices in a partition.

When two moves have the same gain, METIS breaks ties using two further criteria: external degree and partition balance. In the case of a tie, it first chooses the move with the highest specific external degree (see Section 3.4.1). Since all vertices within a coarse vertex are assigned to the same partition upon projection, a move toward a partition with high external degree may reduce volume at finer levels. If instead two candidates have the same gain and the same external degree, the move to the more underweight partition is chosen.

Consider a vertex i being visited in the boundary set during refinement. Algorithm 3 shows how METIS chooses vertex moves using the logic described above.

Algorithm 3 Total communication volume refinement

```
Require: Vertex i with neighbor partitions N_i[0 ... n−1]
  if w(P[i]) − w(i) ≥ W^min then
      for k = n−1 downto 0 do                          ▷ Find first feasible candidate
          if G_i[N_i[k]] ≥ 0 AND
             w(N_i[k]) + w(i) ≤ W^max + f · G_i[N_i[k]] then
              break
          end if
      end for
      if k < 0 then continue to next vertex
      end if
      for j = k−1 downto 0 do                          ▷ Search for a better candidate
          p_j ← N_i[j], p_k ← N_i[k]
          if G_i(p_j) > G_i(p_k) AND
             w(p_j) + w(i) ≤ W^max + f · (G_i(p_j)) then
              k ← j                                    ▷ Better gain, relaxed balance
          else if G_i(p_j) = G_i(p_k) AND
                  ED_{p_j}[i] > ED_{p_k}[i] AND
                  w(p_j) + w(i) ≤ W^max then
              k ← j                                    ▷ Same gain, better connectivity
          else if G_i(p_j) = G_i(p_k) AND
                  ED_{p_j}[i] = ED_{p_k}[i] AND
                  w(p_j) < w(p_k) then
              k ← j                                    ▷ All else equal, prefer underloaded partition
          end if
      end for
      Move i to N_i[k]
  end if
```

5.2 Refinement using partition-specific communication volume

We now describe how refinement is performed using the partition-specific communication volume objective in NVOL. In Section 4.5.2 we explained how, for a boundary vertex i, the gain table SG_i is computed. To briefly recall, each entry $SG_i[p][j]$ records the change in communication volume that partition p would experience if vertex i were moved to neighboring partition j. The rows of the table correspond to affected partitions, while the columns correspond to possible destinations. The goal of NVOL is to minimize the maximum communication volume held by any single partition.

5.2.1 Move selection

During refinement, NVOL evaluates every candidate destination for a boundary vertex by inspecting the gain table. For each potential move of vertex i to a destination partition j, NVOL determines what the highest volume among all affected partitions would be after the move. If any partition would exceed the current global maximum partition volume, the highest volume held by any partition in the entire graph, the move is immediately discarded. As with the global communication volume approach, NVOL also enforces partition weight constraints: any move must keep the destination partition's weight within the allowed bounds.

NVOL ranks candidate moves according to three criteria, applied in the following order of priority: maximum local partition volume, external degree and balance. The first criterion seeks a move that minimizes the local top volume. Out of all the affected partitions, NVOL will choose the move that decreases the top volume of these partitions the most, as long as that top volume is lower than the global maximum. The second criterion is used when two moves have the same local top volume. As a tiebreaker, NVOL favors destinations toward which the considered vertex has a high external degree. This follows the same logic as the total communication volume approach, aiding moves where the vertex is strongly connected. The final criterion is used when the previous two are tied. NVOL prefers moving vertices to lighter partitions to better balance the weight distribution between the partitions.

5.2.2 Encouraging more moves

A deliberate design choice in NVOL was to promote many vertex moves. Especially when choosing the first candidate, NVOL does not require an immediate improvement of the local top volume. Instead, it chooses any move that leads to a local top that is below the global maximum. This means that in practice NVOL might choose moves that increase the local top volume as long as it is below the global maximum.

The goal with this design is to increase the number of moves NVOL performs during a refinement stage. During testing, we experimented with a stricter criterion that required every accepted move to improve the local top volume. Under this stricter rule, significantly fewer vertices were moved during refinement. On average, the stricter rule performed half as many moves as the more relaxed approach. By encouraging more moves, we encountered two effects. First, NVOL was able to explore more solutions: since more moves were performed, more setups were evaluated. Second, we increased the chance of avoiding getting stuck in a local maximum. If no move could possibly improve the partition's volume, it would remain stuck; by allowing a "worsening" move, we increase the chance of finding better solutions. Since the condition still guarantees that the global maximum is never exceeded, the objective value is protected throughout the process.

To prevent this relaxed condition from accepting moves that are too damaging, NVOL incorporates the fudge factor as a safety mechanism. The fudge factor is used to adjust the maximum allowed partition weight; however, in NVOL we multiply it by the gain for the top partition. Specifically, for a vertex i, a partition m with the highest local volume, and a destination partition j, we multiply the fudge factor by $SG_i[m][j]$. This ensures that moves that have a negative impact on the local top partition are more likely to be filtered out by the weight constraint. By making the maximum allowed weight smaller, it gets harder to accept a move. In the opposite case, it works as a relaxing factor, allowing moves that positively improve the local top. In this way, f acts as a soft penalty that discourages harmful moves without outright forbidding all non-improving ones. In contrast to the total volume approach, this is used for all criteria other than balance.

5.2.3 Implementation of refinement

For a boundary vertex i, NVOL iterates over the neighboring partitions N_i. For each candidate destination j, it first determines which of the affected partitions would hold the highest volume after the move. In order to do that, it scans all entries $SG_i[k][j]$ for $k \in N_i$ and computes $PV[k] + SG_i[k][j]$, where $PV[k]$ is the volume partition k holds

before the move. The partition achieving the highest post-move volume is recorded as the local top partition, with volume tmpVol and partition id tmpPid.

If no candidate has been found yet, NVOL applies the relaxed first condition: destination j is accepted as a candidate if tmpVol does not exceed the global maximum partition volume topVol, and the weight constraint $w(j) + w(i) \leq W^{\max} + f \cdot SG_i[\text{tmpPid}][j]$ is satisfied.

Once a first candidate has been found, NVOL tries to find a better destination. A candidate is replaced if the new destination leads to a lower local top volume, using the same weight constraint as above. In the case of a tie, destinations with high external degree $ED_j[i]$ are preferred. If both local top volume and external degree are tied, the destination with the lower current weight is chosen, favoring underloaded partitions. After all neighboring partitions have been considered, vertex i is moved to the final selected destination. The full procedure is given in Algorithm 4.

Algorithm 4 NVOL refinement

```
Require: Vertex i with neighbor partitions N_i[0...n−1];Max partition volume topVol
  if w(P[i]) − w(i) ≥ W^min then
      candidateVol ← −1
      destination ← −1
      for all j ∈ N_i do
          tmpVol ← 0
          tmpPid ← −1
          for all k ∈ N_i do
              if tmpVol< PV[k] + SG_i[k][j] then
                  tmpVol ← PV[k] + SG_i[k][j]
                  tmpPid ← k
              end if
          end for
          if candidateVol = -1 AND
             topVol ≥ tmpVol AND
             w(j) + w(i) ≤ W^max + f · SG_i[tmpPid][j] then
              destination ← j            ▷ Move does not increase max partition volume
              candidateVol ← tmpVol
          else if candidateVol != -1 AND
                  candidateVol > tmpVol AND
                  w(j) + w(i) ≤ W^max + f · SG_i[tmpPid][j] then
              destination ← j            ▷ Lower local max partition volume after move
              candidateVol ← tmpVol
          else if candidateVol != -1 AND
                  candidateVol = tmpVol AND
                  ED_j[i] > ED_destination[i] AND
                  w(j) + w(i) ≤ W^max + f · SG_i[tmpPid][j] then
              destination ← j            ▷ Same gain, better connectivity
              candidateVol ← tmpVol
          else if candidateVol != -1 AND
                  candidateVol = tmpVol AND
                  ED_j[i] = ED_destination[i] AND
                  w(j) < w(destination) then
              destination ← j            ▷ All else equal, prefer underloaded partition
              candidateVol ← tmpVol
          end if
      end for
      Move i to destination
  end if
```

	NVOL02	NVOL07	NVOL05	NVOL08	NVOL08c
heart01	-0.2	-9.4	**-9.5**	-0.2	-8.6
heart02	+8.2	+9.2	+8.9	**+6.8**	+6.5
heart03	-2.7	-2.3	**-3**	-1	-1.7
heart04	+11.4	+10.9	+11.8	+7.8	+8.3
heart05	**+0.8**	+1.0	+1	+1.0	+0.9
heart06	-1.1	-1.0	**-1.2**	-1.1	-1.0
heart07	-3.6	-3.6	-3.5	**-3.9**	-2.9
channel-500x100x100-b050	-10.8	-19.8	-18.8	**-21.0**	-20.5
fe_tooth	-6.7	**-7.9**	-7.6	-5.4	-5.5
hugetrace-00020	-2.5			-2.6	+1.3
hugetric-00020	-1.9	-1.9	-2.0	-2.0	+0.3
hugetric-00000	-5.7	-5.6	-6.0	**-6.1**	+0.5
citationCiteseer	+3.5			+12.0	-2.4
italy_osm	0.0				
europe_osm	0.0				
roadNet-CA	**-0.4**	**-0.4**	**-0.4**	-0.3	-0.3
kmer_U1a	-2.4				
kmer_V2a	0.0				
pa2010	0.0			-0.5	-0.2
coPapersDBLP	+4.0			+19.2	

Chapter 6

Results

Chapter 7

Future Work

Chapter 8

Conclusions

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